Negative-Energy Modalities in Laboratory Quantum Field Systems: Boundary, State, Tensor, and Readout Modalities

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Abstract

Laboratory negative-energy work spans Casimir boundary response, prepared stress-energy states, tensor and averaged energy-condition statements, and instrument-native readouts. This paper supplies the central modality map for a three-paper ensemble built around the White et al. sphere–cylinder Casimir result. The map separates boundary response, state response, tensor/averaging relevance, and readout observability.

The boundary-conditioned modality concerns vacuum response organized by material boundaries, geometry, and surface preparation. The state-conditioned modality concerns negative stress-energy created or selected by field-state preparation, measurement, squeezing, or quantum-energy-teleportation-type protocols. The tensor/averaging modality concerns the stress-energy object: component, contraction, sampling domain, averaging condition, and source comparison. The readout-conditioned modality concerns the apparatus channel: force, pressure, force gradient, resonant frequency, phase, loss, current, voltage, or detector response, together with the material and electromagnetic controls that make the measurement interpretable.

The interface between modalities is the paper's main object. A boundary morphology gains tensor content through stress components, normalization, material modeling, and smearing. A tensor object gains source relevance through a declared source demand and scale comparison. A boundary or state-conditioned object gains laboratory content through an observable, transfer function, controls, nuisance templates, and false-positive logic. The White sphere–cylinder case is used as the worked boundary example: its shell-like morphology, source-scale accounting, and physical readout routes are connected layers with distinct evidentiary roles. The companion audit applies this map to the White geometry; the companion readout paper applies it to pressure, cavity, and superconducting measurement ensembles.

1. PURPOSE

Laboratory negative-energy work already spans several mature strands of physics. Casimir systems supply boundary-conditioned vacuum response. Squeezed, measurement-conditioned, and quantum-energy-teleportation protocols supply state-conditioned stress-energy structures. Energy conditions and quantum energy inequalities supply tensor, averaging, duration, and smearing context. Precision instruments supply force, pressure, frequency, phase, loss, current, voltage, and detector observables.

This paper is the central map for a three-paper ensemble around the White, Vera, Han, Bruccoleri, and MacArthur sphere–cylinder Casimir result [1]. The map separates four modalities: boundary-conditioned response, state-conditioned response, tensor/averaging relevance, and readout-conditioned observability. The separation gives each layer its own evidence while keeping the interfaces visible.

The White sphere–cylinder case is the motivating boundary example. White et al. used worldline numerics on custom Casimir cavities and reported a three-dimensional sphere–cylinder energy-density distribution correlated with the Alcubierre metric source shape [1]. Reading that result field-wise brings together worldline methods for Casimir geometries [2], energy conditions and quantum inequalities [3, 4], precision Casimir metrology [9, 8], and emerging superconducting Casimir sensing [12].

The paper pair that follows this central map uses the same separation. The companion audit asks what the White boundary geometry contributes at the morphology, tensor/source, scale, material, and readout-accounting layers. The companion readout paper asks how the shell-coded object enters pressure, cavity, and superconducting measurement ensembles. This paper supplies the shared modality language for those two more concrete studies.

2. FOUR MODALITIES

A modality is the physical route by which a negative-energy statement is made specific. It records the preparation or constraint, the stress-energy object, the spatial and temporal support, the compensation structure, the readout channel, and the inference supported by the data. Table 1 gives the working classification used in this ensemble.

Modality	Primary object	Supported inference
Boundary-conditioned	Vacuum response fixed by material boundaries, geometry, and field boundary conditions	A finite apparatus geometry organizes Casimir energy, pressure, stress, or a proxy density into a spatial pattern.
State-conditioned	Stress-energy in a prepared, post-selected, squeezed, measured, or feed-forward-conditioned quantum field state	A field state carries a local or smeared stress-energy deficit with a defined preparation record and compensation structure.
Tensor/averaging	Renormalized stress tensor, null or timelike contraction, smeared integral, or energy-condition quantity	The signed object is tied to a specified component, averaging window, source comparison, and quantum-inequality or averaged-condition context.
Readout-conditioned	Measured apparatus response: force, pressure, gradient, frequency, phase, loss, current, voltage, transition, or detector signal	A laboratory observable contains a recoverable coefficient with declared controls, nuisance templates, calibration, and false-positive logic.

Table 1: Four laboratory modalities for negative-energy claims. The modalities can be combined, and each supports a different kind of conclusion.

The same experiment can occupy several modalities. A structured Casimir cavity can be boundary-conditioned in its source of vacuum response, tensor/averaging conditioned in its comparison with $T_{\mu\nu}$, and readout-conditioned in the force or resonator signal used to measure it. A quantum-energy-teleportation protocol can be state-conditioned in its preparation, tensor/averaging conditioned in its local deficit, and readout-conditioned in the detector or work-extraction channel used to score the event [5].

The classification is an interface map. It keeps the paper pair oriented: which object is being advanced, which object is being measured, and which object is being left to a companion calculation or experiment.

3. BOUNDARY-CONDITIONED RESPONSE

The boundary-conditioned modality begins with a material or geometric constraint on quantum fields. Parallel plates, sphere–plate systems, microstructured surfaces, deep trenches, nanomembranes, and superconducting microplates all fit this class. The signed object is a boundary-conditioned change in vacuum energy, stress, pressure, force, or a calculational proxy for those quantities. The control variables are geometry, material response, and surface preparation.

This modality is the native home of Casimir physics. Precision measurements typically recover force, force gradient, pressure, or resonator response, with surface potentials, roughness, edge effects, finite conductivity, temperature, alignment, and material optical data treated as part of the measurement model [9, 10, 11]. MEMS/NEMS and nanomembrane platforms make the same point experimentally: the observable is an apparatus response whose interpretation depends on careful material and electrostatic accounting [6, 7, 8].

Worldline numerics adds a valuable finite-geometry language to this modality. White et al. used worldline numerics to study custom Casimir cavities [1]; Schafer, Huet, and Gies develop the corresponding worldline machinery for fluctuation-induced energy-momentum tensors in Casimir geometries [2]. That combination makes it natural to ask shape questions: where the response concentrates, how it shifts under geometry perturbations, how much signed weight lies in a chosen region, and whether the strongest response tracks a designed boundary feature.

The White sphere–cylinder example is boundary-conditioned at this level. White et al. report that a $1\,\mu$m diameter sphere centered in a nominal $4\,\mu$m diameter cylinder gives a three-dimensional Casimir energy-density distribution correlated with the Alcubierre source shape [1]. The field-facing boundary question is therefore concrete: how does a finite sphere–wall geometry organize the vacuum response, and how much of that organization belongs to an annular shell role?

4. STATE-CONDITIONED RESPONSE

The state-conditioned modality begins with a quantum field state and a preparation record. The signed object is a local or smeared stress-energy feature of that state. Examples include squeezed-state stress-energy profiles, measurement-conditioned field states, and quantum-energy-teleportation-type protocols in which local operations and classical records create a negative stress-energy density in a target region [3, 5].

The preparation record is part of the claim. It states how the field state was made, which degrees of freedom were measured or driven, how the classical record enters, where compensating positive energy appears, and what time window carries the deficit. This modality is dynamic and operational. Its strength is that the sign can be tied to a controllable state-preparation procedure alongside fixed-boundary inference.

State-conditioned negative energy also naturally meets quantum energy inequalities. Ford's review emphasizes the basic QFT fact that local negative energy densities occur, with magnitude and duration governed by quantum constraints [3]. Kontou and Sanders place that fact inside the broader energy condition landscape: quantum fields motivate a refined move from pointwise conditions to averaged and quantum-inequality statements [4]. For state-conditioned claims, duration, smearing, compensation, and averaging are part of the observable content.

This modality sits beside boundary-conditioned Casimir work with a different preparation logic. A Casimir geometry changes the vacuum state through boundary conditions. A QET-style or squeezed-state protocol changes, selects, or conditions the field state through operations. Both can produce negative-stress-energy language, with distinct compensation and readout structures.

5. TENSOR AND AVERAGING RELEVANCE

The tensor/averaging modality asks which stress-energy object is being used. For a relativistic source claim, the object is a component or contraction of the renormalized stress tensor, such as T_{00}, $T_{ab}k^a k^b$, or an integral of T_{ab} over a specified sampling domain. The tensor object is then compared with the source role under discussion.

This modality supplies the bridge from laboratory QFT to gravity-facing language. Energy conditions in general relativity were historically formulated as restrictions on stress-energy; quantum field theory gives well-understood violations of pointwise versions and motivates averaged or quantum-inequality replacements [4]. A laboratory negative-energy measurement can therefore be precise about its level:

- local energy density or pressure in a material/boundary setup;
- a stress-tensor component or finite-difference proxy;
- a null or timelike contraction;
- a smeared or averaged quantity;
- a source-demand comparison for a specified metric model.

This layer is where morphology and source relevance separate cleanly. A boundary-conditioned morphology can be real and reproducible while its calibrated stress scale remains Casimir-scale relative to a gravity-facing source demand. A state-conditioned deficit can be operationally generated while its averaged integrals obey quantum-inequality restrictions. A readout can recover a geometry-locked coefficient while the corresponding tensor component remains an ordinary Casimir-scale object.

The positive role of the tensor/averaging modality is to keep each of those statements useful. Morphology, readout, and source comparison can each carry their own evidence. Tensor and averaging evidence then provide the bridge between them.

6. READOUT-CONDITIONED OBSERVABILITY

The readout-conditioned modality begins with an instrument. Casimir experiments measure force, force gradient, pressure, displacement, oscillator-frequency shift, phase, quality factor, transition behavior, or other apparatus responses. Those responses are sensitive to vacuum-boundary physics through a transduction chain, and the same chain also carries ordinary material and electromagnetic effects.

This modality is where laboratory claims become data statements. A readout statement identifies:

- the observable and units;
- the geometry schedule or state-preparation schedule;
- calibration channels and null controls;
- material, patch, thermal, vibration, alignment, and readout-chain templates;
- the recovery statistic and false-positive test.

Modern Casimir metrology already works in this style. Bimonte et al. combine micromechanical torsional-oscillator measurements with explicit roughness, edge, patch-potential, and optical-data modeling over 0.2–8 μm separations [9]. Patch-potential modeling and Kelvin-probe measurements are central to high-quality force interpretation [10, 11]. Superconducting-transition Casimir sensing adds a cryogenic readout frontier in which tiny pressure changes are compared against electrostatic, thermal, material, and alignment effects [12].

Readout-conditioned observability is therefore a positive experimental language. It gives a boundary or state result an instrument-native test. Pressure, frequency shift, phase, loss, current, and voltage each carry a distinct transfer function, nuisance structure, and claim boundary.

7. INTERFACES BETWEEN MODALITIES

The interesting part of the map is the interface between modalities. A boundary calculation can motivate an instrument. A state-preparation protocol can predict a stress-tensor deficit. A readout experiment can recover a coefficient that is compared with a tensor model. The transfer is scientifically productive when the new modality brings its own evidence.

Table 2 lists common interfaces and the added evidence that makes the handoff meaningful.

Interface	Added evidence	Resulting statement
Boundary morphology to tensor object	Stress-tensor component, normalization, convergence, material model, and smearing convention	The shape is associated with a quantified energy, pressure, or stress channel.
Tensor object to source comparison	Metric source demand, units, sign convention, averaging window, and scale ratio	The laboratory stress-energy object can be compared with a declared gravity-facing source role.
Boundary or state object to readout	Observable model, transfer function, controls, nuisance templates, and false-positive test	The apparatus can search for the corresponding coefficient in measured data.
Readout coefficient to field interpretation	Calibration closure, background separation, phase/geometry locking, and replication under controls	The measured signal can be assigned to the boundary or state-conditioned field object under the declared model.
State preparation to averaged-energy statement	Sampling function, duration, compensating positive energy, and quantum inequality comparison	The local deficit is placed inside a finite-time and finite-region energy ledger.

Table 2: Evidence added when a result moves across modalities. The resulting statement names the new physical object being tested or compared.

This interface table is the practical core of the modality map. A morphology result can stand as a morphology result. A readout architecture can stand as a readout architecture. A source comparison can stand as a source comparison. The connecting evidence shows how those results meet.

8. WHITE SPHERE–CYLINDER CASE STUDY

The White sphere–cylinder result is a useful case study because it touches several modalities at once. The original paper reports a boundary-conditioned worldline-numerics result: a custom Casimir geometry whose sphere–cylinder toy model is qualitatively correlated with the Alcubierre energy-density shape [1]. That visual resemblance brings in the tensor/source modality. The proposed voltage, current, photon, or electron detection ideas bring in the readout modality. The case is therefore naturally read as a modality-interface problem.

The companion audit in this paper ensemble treats the White geometry as a boundary-conditioned and tensor/source case. Its role-resolved question is: what remains after separating the sphere, cylinder wall, annular shell, central readout path, far field, material/background accounting, and source scale? Within that structure, the shell-like boundary morphology can be supported while direct gravity-facing scale and central timing/readout routes are bounded by calibration and recovery gates.

The companion readout paper treats the shell-coded object as a readout-conditioned target. Its question is: which physical apparatus ensembles could test a geometry-locked boundary response? Pressure cells, high-Q cavities, and superconducting resonators become branch-native readouts. Their success criteria are pressure contrast, eigenmode shift, phase/loss response, or cryogenic microwave response with material and electromagnetic controls.

Placed in the modality map, the case study has a clean identity:

- Boundary modality: the finite sphere–cylinder geometry organizes a shell-like Casimir response.
- Tensor/averaging modality: stress scale, source demand, and averaging decide the gravity-facing interpretation.
- Readout modality: pressure, cavity, superconducting, voltage, current, photon, or electron channels decide what an apparatus can recover.
- State modality: the White setup is primarily boundary-conditioned; a state-conditioned extension would add a separate preparation protocol.

This is the value of the central paper. It lets the White pair stay technically sharp while the ensemble has a shared modality map. It also lets other negative-energy experiments locate themselves on the same map.

9. USING THE MODALITY MAP

For this ensemble, the map reduces each negative-energy statement to six coordinates:

1. **Modality.** Is the sign boundary-conditioned, state-conditioned, tensor/averaging-conditioned, readout-conditioned, or a declared combination?
2. **Object.** Is the object energy, pressure, force, stress, frequency shift, phase, loss, current, voltage, detector response, or a model coefficient?
3. **Support.** Where and when does the signed object live, and how is the support defined?
4. **Compensation.** Where does the associated positive energy, material response, reset cost, or background response appear?
5. **Controls.** Which material, electromagnetic, thermal, patch, alignment, drift, and readout-chain controls share the same apparatus?
6. **Interface path.** Which additional evidence would connect the result to another modality?

Table 3 summarizes the resulting evidence layers.

Evidence layer	Evidence basis	Resulting statement
Geometry organizes response	Finite-boundary calculation, perturbation checks, morphology metrics	The apparatus geometry carries a reproducible boundary-conditioned pattern.
Stress channel quantified	Stress tensor or calibrated proxy, material model, scale estimate	The pattern has a quantified energy, pressure, or stress role.
Operational state deficit	Preparation record, detector/work ledger, duration and compensation	The field state carries a controlled local or smeared deficit.
Instrument recovers coefficient	Measured observable, controls, template separation, false-positive test	The apparatus sees a recoverable boundary or state-conditioned coefficient.
Source relevance established	Tensor component, averaging, source-demand comparison, scale closure	The stress-energy object can be assigned to the declared gravity-facing source role.

Table 3: Evidence layers enabled by the modality map. Each layer is useful on its own and can be connected to another layer by adding the evidence named in the interface table.

This map keeps the ensemble's claims at the right level: boundary morphology, field-state preparation, stress-tensor scale, readout recovery, and source relevance can be discussed as connected but distinct evidence layers.

10. IMPLICATIONS FOR LABORATORY PROGRAMS

The modality map suggests a constructive path for laboratory QFT programs. One starts from the modality with the strongest existing evidence. For many Casimir platforms, that is boundary-conditioned response plus readout-conditioned force or pressure metrology. For QET-style protocols, it is state-conditioned preparation plus operational energy extraction or detector response. For energy-condition studies, it is tensor/averaging structure plus duration and smearing constraints.

The next step is to design the interface deliberately. A finite-boundary morphology gains an apparatus role through a stress or readout model. A readout signal gains a field role through template separation and calibration. A local stress-energy deficit gains an averaged-energy role through smearing and compensation. A gravity-facing source comparison gains content through calibrated stress scale, tensor component, sign convention, and source-demand ratio.

Ordinary apparatus physics is part of the field experiment. Material response, patch potentials, finite conductivity, thermal behavior, roughness, mode mixing, vibration, alignment, and readout-chain effects are the channels through which laboratory negative-energy claims become precise. In high-quality Casimir work, those channels already provide the calibration structure that makes the measurement scientific [9, 10, 11, 12].

This has a positive consequence for speculative-adjacent systems. A visually suggestive negative-energy morphology can be valuable at an early source-assignment stage. It can be a boundary-conditioned target, a stress-channel hypothesis, a readout template, or a discrimination schedule. Each role supports real progress when it is named.

11. REPRODUCIBILITY AND SOURCE BASIS

The cited source PDFs for this central paper are mirrored in the local `white_casimir_intersection/sources` directory. That directory includes the White et al. EPJ C paper, QFT negative-energy and energy-condition references, the worldline stress-tensor reference, engineered negative stress-energy via quantum energy teleportation, and the Casimir metrology, patch-potential, and superconducting-transition references used for the readout discussion.

This manuscript is a conceptual synthesis. Its reproducible content is the modality map: the modality definitions, interface-evidence table, and evidence-layer table. The two companion papers supply the worked White sphere–cylinder audit and the physical readout architecture application.

12. CONCLUSION

The modality map gives this ensemble a stable center. It treats four connected laboratory objects: boundary response, prepared-state response, tensor/averaging relevance, and readout observability. Their interfaces are the places where a negative-energy result becomes experimentally and theoretically informative.

The White sphere–cylinder case illustrates the map in a concrete setting. The reported shell-like Casimir boundary morphology, the stress/source scale, and the proposed readout routes belong to connected layers. The companion audit uses the map to keep boundary morphology, source scale, and central readout accounting distinct. The companion readout paper uses the same map to ask how pressure, cavity, and superconducting instruments could test the shell-coded boundary response.

The field-facing result is a compact modality map for laboratory QFT. It names the object carrying the sign, the support assigned to that object, the compensation or background structure around it, the apparatus controls that make the readout meaningful, and the additional evidence that connects one modality to another. Boundary effects, prepared field states, stress-tensor constraints, and instrument responses remain scientifically connected while retaining their specific meanings.

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